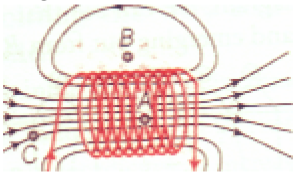
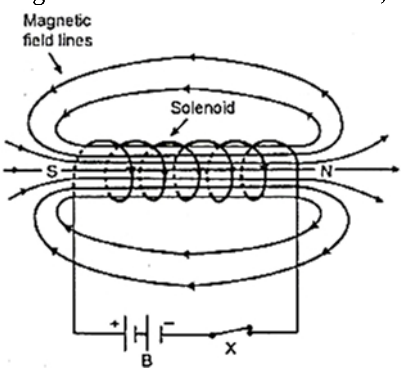
Solution

MAGNETISM -06

Class 10 - Science

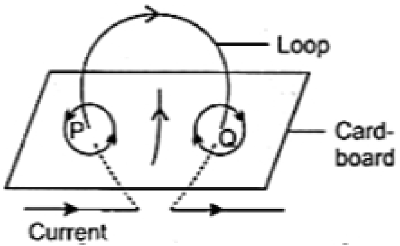
1. Earth wire is used as a safety measure especially for electric appliances having a metallic body. The metallic body of appliances like electric press, fans, toaster, refrigerators, etc are connected to earth wire which provides an easy path for current to go to the earth in case live wire touches the body of appliance incidentally. The user will not suffer a severe electric shock even of touching a defective appliance.
2. Magnetic field lines due to a solenoid


In case of an ideal solenoid, magnetic field strength is maximum at point A and is minimum or zero at point B. This is because the magnetic field is strong, where magnetic field lines are far apart. At the point, C, the density of the field lines is less than that of point A, but greater than that of point B. So, the order of the magnetic field at points A, B and C is $B_B < B_C < B_A$.
3. In case of movement of a charged particle, the magnetic field is created around the path on which the charged particle moves
   i. Yes, because α-particles are charged particles so when they move they create a magnetic field around it.
   ii. No, as neutrons carry zero charges, so no magnetic field would be produced around its path.
4. **Short circuiting :** If sometimes a live wire touches neutral wire or earth wire, a large current flows through the circuit due to almost zero resistance of the circuit. This is called short-circuiting. To save the circuit from damage feared due to over-loading a short-circuiting, a fuse of proper rating is put in each circuit.
Over loading : The supply wires as well the wires used in household wiring has a specific rating. The rating of 15 A means that if a current upto 15 A is passed through circuit, there is no likely damaged feared to the circuit. But if current more than maximum allowed limit is passed, there may be excessive heating of the wires and it may damage the wiring due to excessive heating.
5. Place a compass needle at the given point. If it stays in the north-south direction, then the magnetic field is due to earth. If the needle points along any direction other than north-south direction, then the field is due to some current carrying conductor.
6. A solenoid is a coil containing many circular turns. These wires are wrapped closely in the shape of a cylinder. When a current is run via a solenoid, it acts like a bar magnet. Its opposite end functions as the magnetic South Pole and its opposite end as the magnetic North Pole. In other words, the field lines appear from one end and merge into another, much like a bar magnet.


7. Whenever there is an electric current, there is a magnetic field. Even the extremely weak ion currents that travel along the nerve cells in our body produce magnetic fields. When we try to touch something, our nerves carry an electric impulse we need to use. This impulse created a temporary magnetic field. These fields are about one billionth as weak as the Earth's field. Two main organs in the human body where the magnetic field produced is significant are heart and brain.
8. A current carrying solenoid behaves like a bar magnet. It rest in geographic north-south direction.
9. **Overloading:** When too many devices are connected to the same circuit and are operated at time, the wire draws a much large current (greater then its capacity) from mains is called overloading. This causes heating in the circuit that can lead to sparking.
Causes of overloading:
   i. Live and neutral wires come in contact with each other overloading occurs.
   ii. Connecting too many appliances to a single socket.
Prevention of overloading
One should always install a fuse or Mini Circuit Breaker to every switchboard to prevent this situation. Proper earthing should be

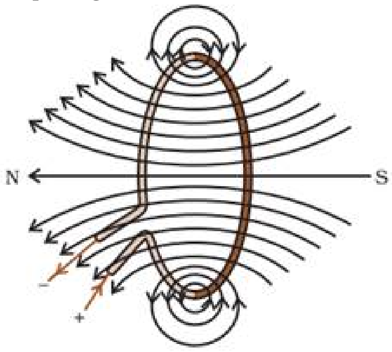
installed in the domestic electric circuit. If the earthing is appropriately done in a circuit, then the excess current will flow through the earthing to prevent the circuit's overload.

10. An electric switch should not be touched with wet hands while putting it on or off because if water reaches the live wire, it forms a conducting layer between the hand and the live wire of the switch due to which a strong current passes through the hand and the user gets an electric shock.
11. We should take following precautions to avoid the overloading of domestic electric circuits:
    a. Two separate circuits should be used, one of 5A current rating of bulbs, fans, tubes etc. and the other 15 A current rating for appliances with higher current rating such as geysers, air coolers, electric iron, electric stoves etc.
    b. Too many appliances should never be connected to a single socket.
    c. A fuse of appropriate current rating should be used with the electric circuit.
12. a. **Fleming's left-hand rule:** Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor.
    b. South
13. The compass needle is small bar magnet. When a compass needle is brought near a bar magnet then due to repulsive force between like poles and attraction between unlike poles, the compass needle is deflected and settle in the direction of net magnetic field.
14. The factors affecting the strength of a magnetic field at a point due to a straight current-carrying conductor are the magnitude of the electric current and the perpendicular distance between the point and the conductor. Right-hand thumb rule determines the direction of magnetic field produced in this case.
15. The S.I unit of magnetic field is Tesla (T). The magnetic field strength is said to be one Tesla if one meter long conductor placed perpendicular to the direction of magnetic field ,carrying one ampere current experiences one newton of force.
16. i. As the distance from conductor increases, the strength of the magnetic field decreases i.e. $B \propto \frac{1}{r}$.
    ii. As the magnitude of the current flowing through the conductor increases, the strength of magnetic field increases, i.e. B $\propto$ I.
17. Figure shows the magnetic lines of force due to current Carrying Coil.

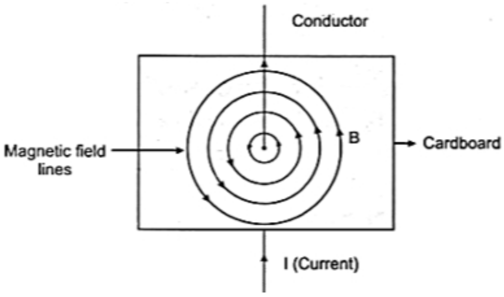


    The magnitude of magnetic field at the centre of coil depend on:
    a. the strength of current in the coil, and
    b. the number of turns in the coil.
18. Force experienced by a current-carrying conductor placed in a magnetic field is maximum (largest) when the field is perpendicular to current-carrying conductor.
19. a. Series, live
    b. thicker
    c. bar magnet
    d. magnetic field
20. At the axis of a current carrying circular loop, magnetic field lines are straight. Hence they appear closer to each other. When we move towards the periphery of current carrying circular loop, magnetic field lines turn so as to encircle the conductor. Due to this, they appear to be diverging.
21. a. Since current at end X is anticlockwise, the polarity at that end is North pole.
    b. Current at end Y is clockwise, hence polarity at that end in South pole.
    c. Clock-face rule is used to determine the polarities of the two faces of a current-carrying circular loop.
22. The lines drawn in a magnetic field along which north magnetic pole moves, are called magnetic field lines. The characteristic properties of magnetic field lines are:
    i. The magnetic lines originate from north pole and ends at south pole.
    ii. The magnetic lines do not intersect each other.

23. i. Divergence or degree of closeness of magnetic field lines near the ends of a current-carrying straight solenoid indicates an increase in the strength of the magnetic field near the ends of the solenoid.
    ii. A current-carrying solenoid acts as a bar magnet. We know that a freely suspended bar magnet aligns itself in the North-South direction. So, a freely suspended current-carrying solenoid also aligns itself in the North-South direction.
    iii. Burnt out fuse cannot be re-used. Also, a fuse wire works because of its lower melting point. If the fuse with a larger rating is used with an appliance, the fuse wire shall not melt and hence would fail to serve the required purpose. So, a new fuse of the same rating should be used for electrical safety.
24. Fleming’s left hand rule states that stretch the forefinger, the central finger and the thumb of your left hand mutually perpendicular to each other. If the forefinger shows the direction of the magnetic field and central finger that of the current, then the thumb will point towards the direction of motion of the conductor.
25. In a current carrying circular loop direction of magnetic field lines can be found by the Right-Hand Thumb Rule. Thumb is showing direction of current and fingers shows the direction of magnetic field lines. This means magnetic field lines are around the conducting wire. But circular shape of the conductor means that field lines at different points of the loop appear to be making many rings around the periphery of the loop. It can be visualized like many small rings looping around the periphery of a big ring.
    Effect of Number of turns in a coil: we know that if number of turns in coil is increased, then magnetic field will also increase. Due to this, strength of magnetic field increases with increased number of turns in a coil. Magnetic field lines through a circular loop are given below:-



26. The magnetic field lines of force are the lines drawn in a magnetic field along which a hypothetical north magnetic pole would move if it is free to do so.
    The direction of a magnetic field at a point is the direction of the resultant force acting on a hypothetical north pole placed at that point.The tangent at any point on the magnetic field line gives the direction of magnetic field at that point. The direction of the magnetic field at a point can be found by placing a small magnetic compass at that point. The north end of the needle indicates the direction of the field.
    Two important properties of the magnetic lines of force are:
    (i) The magnetic lines of force start from north pole and terminate at south pole. Inside the magnet they travel from south pole to north pole. Thus, they are closed curves.
    (ii) They do not intersect each other because at the point of intersection there will be two directions of same magnetic field which is impossible.
27. **Three methods of producing magnetic fields are as follows:**
    (a) Magnetic field can be produced by placing a permanent magnet or a horse-shoe magnet at the place, where magnetic field is required.
    (b) Magnetic field is produced around a current carrying straight conductor or a current carrying coil.
    (c) A very good method to produce magnetic field is due to flow of current in a solenoid.
28. (1) Magnetic field around a magnet.
    (2) Magnetic field around current carrying conductor.
    (3) Magnetic field around a current carrying solenoid.
    (4) Magnetic field around an electromagnet.
29. When an electric current is passed through a metallic conductor, a magnetic field is set-up around it which exists so long as the current flows in the conductor. This is called magnetic effect of electric current.
30. i. Fleming's left hand rule is used to find the direction of force acting on a current carrying conductor, placed in a magnetic field.
    ii. In case (i) the electron is traveling perpendicular to the field and in case (iii) it is traveling parallel to the field. Hence the force is maximum and minimum on the electron in each case.

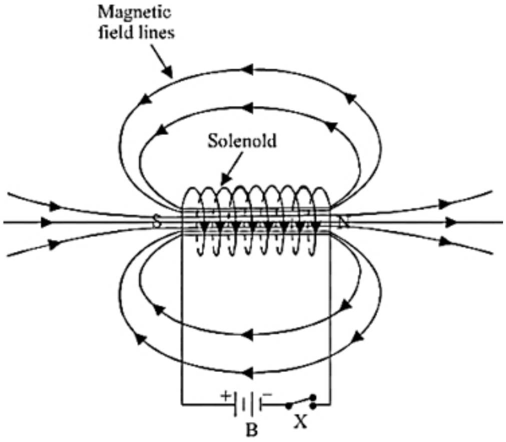
31. a. 

b. Right hand thumb rule.

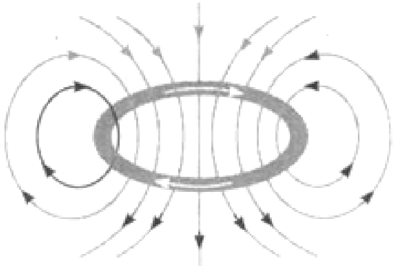
32. A uniform magnetic field in a region is represented by drawing parallel and equidistant straight lines, all pointing in the same direction as shown in figure:

33. Strength of the magnetic field falls as distance increases. This is indicated by the decrease in degree of closeness of the lines of field.

34. A solenoid is a long coil containing a large number of close turns of insulated copper wire. Magnetic field lines inside a current-carrying solenoid are uniform.



35. Since the current passes through the loop in clockwise direction, therefore, the front face of the loop will be the south pole and the back face, i.e., the face touching the table will be north pole. According to right-hand rule, the direction of magnetic field inside the loop will be pointing downward. Outside the loop, the direction of the magnetic field will be upward.



36. It is more dangerous to touch the live wire rather than the neutral wire because live wire has a high potential of 220 V, whereas neutral wire has zero potential.

37. Total power of appliances used simultaneously = (3 $\times$ 100) + (8 $\times$ 40)

= 300 + 320

= 620 W

Voltage of mains, V = 220 V

Current drawn from mains, I = $\frac{P}{V}$ = $\frac{620}{220}$ = 2.82 A

Excess current available = 5A - 2.82 A = 2.18 A

Current drawn by each of 60 W bulb at 220 V,

I = $\frac{P}{V}$ = $\frac{60}{220}$ = 0.27A

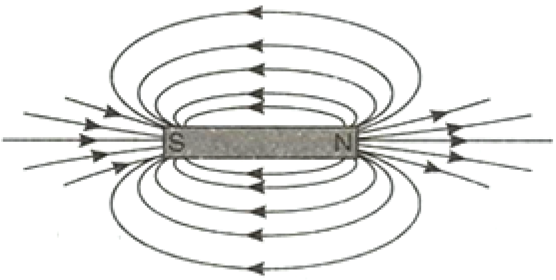
Number of additional bulbs of 60 W which can be lighted,

n = $\frac{\text{Excess current available}}{\text{Current drawn by each bulb}}$

= $\frac{2.18}{0.27}$ = 8.07

Therefore, eight more 60 W bulbs can be lighted.

38. If sometimes a live wire touches neutral wire or earth wire, a large current flows through the circuit due to almost zero resistance of the circuit. This is called short-circuiting. To save the circuit from damage feared due to over-loading or short circuiting, a fuse of proper rating is put in each circuit.

39. 

40. **(a)** The maximum amount of current that can be passed through the fuse wire without melting it.
(b) Copper or alloy of lead and tin.
(c) Electric fuses and earth wire.

41. If either the insulation of wires used in an electric circuit is damaged or there is a fault in the appliances, live wire and neutral wire may come in direct contact. As a result, the current in the circuit abruptly rises and short circuiting occurs.

42. Power, P = 2 kW = 2 $\times$ 1000 = 2000 W
Voltage supply, V = 220 volt
Current drawn by the appliance, I = $\frac{P}{V}$ = $\frac{2000}{220}$ = 9.1 A
No, the 8 A fuse cannot be used, because it will blow off as soon as the appliance is switched on.

43.

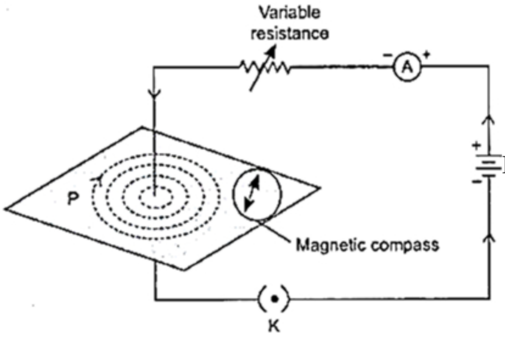
Permanent Magnet	Electromagnet
(i) Very strong electromagnets cannot be produced.	(i) Very strong electromagnets can be produced.
(ii) Its strength is fixed.	(ii) Strength can be changed by changing the current through coil.
(iii) Polarities are fixed.	(iii) Polarities can be reversed by changing the direction of the current.
(iv) Cannot be immediately demagnetized.	(iv) Can be demagnetized immediately by stopping the current in the coil.
(v)Example of permanent magnet is a Bar Magnet	(v) Example of a temporary magnet is solenoid wounded across a nail and connected to a battery.

44. i. The magnitude of force acting on a current carrying conductor placed in a magnetic field will be zero, when the current carrying conductor is in the direction of magnetic field.
ii. The magnitude of force acting on a current carrying conductor placed in a magnetic field will be maximum, when the current carrying conductor is normal (perpendicular) to the magnetic field.

45. Electric short-circuit occurs when :
(a) live wire incidentally touches neutral or earth wire
(b) Insulation gets hardened by the excessive use.
(c) current passed through wire is more than its rating.
(d) insulation around the current carrying wires is weak.

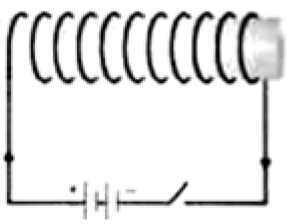
46. The earth pin is made long so that the earth connect is made first and this ensures the safety of the user. If the appliance is defective, then as soon as the live pin gets connected, a strong current flow through the earth wire and the fuse blows off. The earth pin is thicker than the other two pins so that even by mistake it cannot be inserted in the hole for the live or neutral connection.

47. Using a compass needle we can determine the magnetic field. When we move away from the compass needle from the straight wire, the deflection of the needle decreases which implies the strength of the magnetic field decreases, as the strength of magnetic field produced by a straight wire at any point is inversely proportional to the distance of the point from the wire.

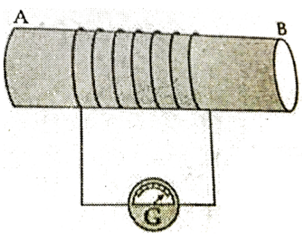


Magnetic field lines produced around a straight current carrying conductor passing through a cardboard.

48. Fuse is used for protecting appliances due to short-circuiting or overloading. The fuse is rated for a certain maximum current and blows off when a current more than the rated value flows through it. If a fuse is replaced by one with larger ratings, the appliances may get damaged while the protecting fuse does not burn off. This practice of using fuse of improper rating should always be avoided.
49. The magnetic field generated in the centre, or core, of a current carrying solenoid is essentially uniform, and is directed along the axis of the solenoid. Outside the solenoid, the magnetic field is far weaker.
50. The magnetic field produced due to current flowing in a coil or a solenoid can be used to magnetise a material like soft iron temporarily. The insulated copper wire is wrapped on a soft iron piece. When current is passed through the coil using a battery and closing a key the iron piece behaves like a bar magnet as long as current is being passed. Such a magnet is called an electromagnet.



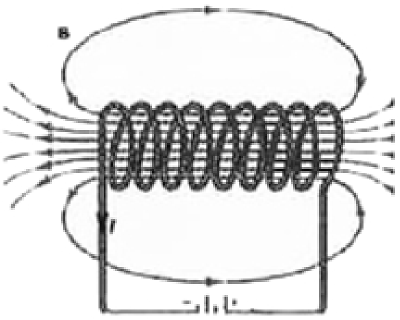
51. The magnetic lines of force due to current in the straight conductor XY are shown in figure given alongside. The arrows on the magnetic lines of force shows the direction of magnetic field. The magnitude of magnetic field at a point depends on:
    i. The strength of current in the conductor, and
    ii. The distance of point from the conductor.
52. i. Right hand thumb rule will determine the direction of a magnetic field around a straight conductor carrying current.
    ii. Fleming's left had rule will determine the direction of a force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.
    iii. Fleming's right hand rule will determine the direction of current induced in a coil due to its rotation in a magnetic field.
53. Power, P = 2 kW
    = 2×1000 W
    = 2000 W
    Voltage, V = 220 V
    Current drawn, I = ?
    Power, P = $V \times I$
    I = $\frac{P}{V}$
    = $\frac{2000}{220}$
    = 9 A(approx.)
    The current drawn by this electric oven is 9 A whereas the fuse in the circuit is only 5 A capacity. When a high current of 9 A flows through the 5 A fuse, the fuse wire will get heated too much, melt and break the circuit. Therefore, when a 2 kW power rating electric oven is operated in a circuit having 5 A fuse, the fuse will blow off cutting off the power supply in this circuit.
54. i. A coil of insulated wire is connected to a galvanometer and if a bar magnet with its south pole towards one face of the coil is



a. moved quickly towards it, the galvanometer is deflected towards the left.
b. moved quickly away from it, the galvanometer is deflected towards the right
c. placed near its one face, the deflection of the galvanometer is zero.

ii. This phenomenon involved in this activity is 'Electromagnetic Induction'.
iii. Motion of a magnet with respect to coil induces an electric current in the coil which lasts so long as the motion is taking place/change in the magnetic field around a coil produces an induced current in it.

55. Power of 1 tube light, P = 40 W
Voltage supply, V = 220 V
Therefore, current drawn by each tube light,
I = $\frac{P}{V}$ = $\frac{40}{220}$ = 0.18 A
Therefore, maximum number of tube lights which can be used,
n = $\frac{\text{Current rating of fuse}}{\text{Current drawn by each tube light}}$
= $\frac{5}{0.18}$ = 27.7
Hence, a maximum number of tube light which can be used = 27.
56. A compass needle is itself a magnet and when placed near a magnet gets deflected due to the replusive magnetic force between like poles and attractive magnetic force between unlike force of the magnet.
57. Clock face rule is used to determine the polarity of the two faces of a current carrying circular loop. According to this rule, "If the current around the face of circular wire flows in the clockwise direction, then that face of the circular wire will be south pole (S-Pole) and if the current around the face of circular wire flows in the anticlockwise direction, then that face of the circular wire will be north pole (N-Pole)."
58. **Magnetism in Human being and Animals :** Whenever there is a flow of an electric current in a wire or solenoid, there is a magnetic field around it. Extremely small current flow along our nerve cells in our body. Whenever we use a sense nerve say we touch a body, our body carry an electric impulse. This produces a very weak magnetic field [About one billionth (10^{-9}th) of that of the earth] This magnetic field produces significant effect in our brain and heart. Magnetism is very useful in obtaining the image of different parts of body by MRI [Magnetic Resonance Imaging]. It has very important uses in medicine also.
59. Suppose x number of such bulbs can be used.
Power of one bulb = 100 watt
Power of x bulbs (p) = 100x watts
Potential difference (V) =220 Volts
I = 5 ampere
We know, P = VI
100x = 220$\times$ 5
Therefore, x = $\frac{220 \times 5}{100}$
= 11 bulbs
60. a. A coil of insulated copper wire wound in the form of a cylinder is a solenoid. When current is passed through a solenoid, it produces magnetic field lines like a bar magnet.
b. The pattern of magnetic field lines of a solenoid is shown in the figure. Inside the solenoid, field lines are parallel to each other This indicates that the magnetic field is uniform .



61. (i) These travel from north pole to south pole outside the magnet and south pole to north pole inside the magnet.
(ii) These are closed and continuous curves.
(iii) Two magnetic field lines never intersect each other.
62. **Three sources of magnetic field are:**
(i) Permanent magnets
(ii) Electromagnets

(iii) Earth's magnetic field.

63. The displacement of the conductor
    (i) will increase on increasing the current
    (ii) Will decrease on using a weak horse shoe magnet.

64. i. The force experienced by a current-carrying straight conductor is maximum when the conductor is placed perpendicular to the direction of magnetic field.
    ii. The force experienced by a current-carrying straight conductor is minimum when the conductor is placed along the direction of magnetic field whether parallel or antiparallel.

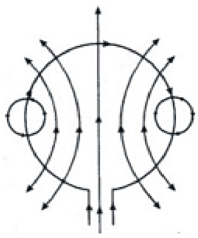
65. Following are the methods of producing magnetic fields:
    i. By using a permanent magnet we can produce a magnetic field and it can be visualized by spreading iron fillings on a white paper and keeping a magnet beneath the paper.
    ii. A current-carrying straight conductor produces a magnetic field.

66. Current-carrying loops behave like bar magnets and both have their associated field lines. This modifies the already existing earth's maernfitic field and a deflection results.

67. **(a)** A short circuit is simply a low resistance connection between the two conductors supplying electrical power to any circuit. Due to this huge amount of current flows through the circuit which is turn produces more heat which can cause fire.
    (b) Overloading means large amount of current flows in the circuit. It can happen when many electrical appliances of high power ratings are connected in a single socket.
    Overloading can be avoided by the following methods:
    (i) Not supplying power to too many appliances through a single socket.
    (ii) To apply preventive methods of short circuiting.

68. The increase in deflection is seen as the strength of magnetic field is directly proportional to the current flowing in the conductor.

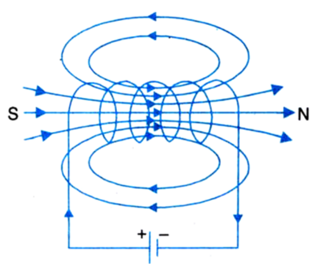
69. The direction of the magnetic field inside and outside is loop as shown.



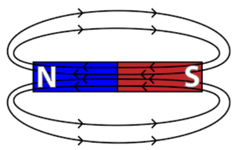
In the following diagram the current is flowing clockwise. If we are applying right-hand thumb rule to the left side of the loop then the direction of magnetic field lines inside the loop are going into the table while outside the loop they are coming out of the table. If we are applying the right-hand thumb rule to the right side of the loop then the direction of magnetic field lines inside the loop are again going into the table while outside the loop they are coming out of the table.

70. a. It is coil containing many circular terns, there wire are wrapped closely in the shape of a cylinder and it is used as an electromagnet.
    It also refers to any device that converts electrical energy to mechanical energy using a solenoid. The device creates a magnetic field from electric current and uses the magnetic field to create linear motion.
    In above image:-
    1) The magnetic field for the current carrying solenoid
    2) For bar magnet.

1. The magnetic field for the current-carrying solenoid



2 For bar magnet.



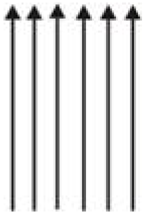
   b. List of the distinguishing features between the two fields:
      i. The poles of the bar magnet do not lie exactly at the end of the magnet but are somewhat inside. In a solenoid, poles can be considered to be lying at the edge.
      ii. The magnetism retains in the bar magnet naturally but in the solenoid, the magnetism is there so long current flows through it.

iii. A magnetic field of a bar magnet emanates from throughout the body of the magnet, with more intensity at the poles. While there is no field emanating from the lateral surface of the solenoid.

71. a. The rods move if a current is sent through them from B to A. The force acting on the current carrying rod is what is responsible for this displacement. The rod is pulled to the left by the magnet, which causes a deflection of the rod in that direction.

b. No change is observed when the axis of the rod "AB" is moved and aligned parallel to the magnetic field and current is passed in the rod in the same direction. Since there is no force, so there will be no motion.

72. Uniform magnetic field is shown by equidistant and parallel lines as shown. The parallel lines are close to each other, if the field is strong. Stronger the field, closer are the lines.



73. The direction in which a compass comes to rest in the magnetic field is called direction of line of force. No two magnetic lines of force cross each other because if they did so, it will mean that at the point of intersection, the compass needle would point towards two directions which is not possible. Hence two magnetic lines never intersect each other.

74. Here, P = 3 kW = 3000 W, V = 220 V Current drawn by motor,

$I = \frac{P}{V}$

$= \frac{3000}{220}$ A

= 13.6 A

Hence, current rating of fuse to be used with this motor is 15 A.

75. **Hazards of electricity :** Electricity is the most important source of energy. Whereas a proper use of this is a boon, improper use is very dangerous and may prove to be fatal. There are number of hazards.

(1) Loose connections, defective switches and sockets can cause sparking which may lead to fire.

(2) If a person touches a live wire, he gets a very sever shock. This may prove fatal.

(3) Short-circuiting due to damaged wires or overloading of circuits can cause fires.